

NovaDrive Phase 2: alternate suppliers, summary of our research

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The short version

We found one real company to replace each of the four riskiest confirmed suppliers, and checked each on six factors using the companies' own public pages. These are shortlisted candidates for sourcing to validate, not final replacements.

Risky supplier (risk %)	What we need to replace	Alternate found
Jade Printed Circuits (69.0%)	Circuit-board substrates for C10 and B10	TTM Technologies
IonPeak Semiconductor (65.3%)	Power semiconductor dies for M10/M20	Infineon
Orion Ceramics (57.5%)	Ceramic substrates for M10/M20	CeramTec
Meridian Dielectrics (57.4%)	Dielectric film for C20 capacitors	Toray

Which suppliers we target, and why

We target the four confirmed suppliers whose "Next step" in the risk scorecard reads "Search alternates now". The scorecard ranks all 24 suppliers on six risk dimensions, and these four combine a high score with solid evidence behind their links.

Supplier	Rank and risk	Why replace it (our data)	Own financial health (our data)	Weeks to qualify
Jade Printed Circuits (Z01)	1st, 69.0%, High confidence	Feeds 3 Tier-1s (Cobalt, Grove, HarborSense). 100% of that component's supply goes through it. DOC-079 says no independent substrate is qualified. All 3 products (USD 1,560m) sit behind it.	Current ratio 1.55, debt/EBITDA 3.1, on-time 86.6% (down 6.4 points since January). Middling, so the risk is position and location.	12
IonPeak Semiconductor (Z01)	4th, 65.3%, High confidence	Only die source behind both power-assembly suppliers (Aster, Boreal). DOC-078 says no second die source is qualified. All 3 products depend on it.	Current ratio 1.8, debt/EBITDA 2.4, on-time 85.2% (down 8.8 points).	16
Orion Ceramics (Z01)	5th, 57.5%, Medium confidence	Ceramic substrates for Aster and Boreal. All 3 products. In the Z01 flood-watch zone. Only one source family backs the link.	Current ratio 2.3, debt/EBITDA 2.7, on-time 82.2% (up 0.7 points).	16 (using the parent component)

Supplier	Rank and risk	Why replace it (our data)	Own financial health (our data)	Weeks to qualify
Meridian Dielectrics (Z03)	6th, 57.4%, Medium confidence	Dielectric film for Delta Capacitor Works, the only C20 supplier. P1 and P2 depend on it: USD 1,144m, 73% of product revenue.	Weakest in the network: current ratio 0.62, debt/EBITDA 7.2, on-time 76.4% (down 17.6 points). Plus active financing pressure (EV-003).	8

Two things to notice. Meridian is in Z03, not the flood zone, so its risk comes from finances. The other three all sit in Z01, where hazard, logistics and infrastructure scores are 82, 76 and 71.

Why not Umber and Verdant (ranks 2 and 3)? Their scores are high mainly because data is missing. Both have Low confidence, and Verdant's link to IonPeak is only inferred. Their next step is to verify first. The same applies to Warden, Xenon, Yarrow and Zenith. Checking them can run alongside the alternate search, and Umber (silicon-carbide wafers) matters when we look at who could replace IonPeak.

How we search for an alternate

We judge every real company on six factors from the case handbook, and we score that fit separately from risk. A good alternate can make the exact item, serves the right industry, has plants in several places, has the scale, has a track record, and can be qualified in reasonable time.

Factor	What we check for each candidate	Anchor from our data
Technical capability	Can they make the exact item: power dies, circuit-board substrates, ceramic substrates, dielectric film?	Component notes in Components Context
Application relevance	Do they already supply industrial drives, EV or industrial charging, or battery systems?	Use case for each component

Factor	What we check for each candidate	Anchor from our data
Manufacturing footprint	Several plants in several regions, and not in the Z01 flood zone	Z01 holds 5 network sites: IonPeak, Jade, Orion, Umber, Verdant
Scale	Can they plausibly handle the volume? Judge from plants, staff, sales and capacity	No supplier spend or volume in the data, so mark this "sourcing to validate"
Demonstrated industry presence	Named customers or products in these industries, with dates	Public web evidence
Qualification	Likely onboarding time and quality certificates	12 weeks for C10 and B10, 16 for M10/M20, 8 for C20

How we read each web page

1. Copy the exact sentence, the link, and the date.
2. Label it "claimed" (the company's own page) or "independent" (a filing, a certificate database, a customer announcement).
3. If a fact isn't on the page, write "not shown". That means ask, not "they don't have it".
4. Check the date, because plant lists change.
5. Don't stretch a claim: "battery management" as an application doesn't prove they can make our exact board.

How we rate. Each factor is Strong, Partial or Not shown. Capability and application relevance must both be at least Partial, or the company drops out. Everything on the web pages is a company claim, so all ratings are claim-level.

Two corrections to keep in mind. USD 1.56B is NovaDrive's product revenue, not what we buy from any supplier, so it isn't a yardstick for supplier scale. And Z01 is a fictional zone, so "outside Z01" can't be tested for a real company. We use plant spread instead.

The four alternates, how they work and how they score

Each alternate would be a second source qualified next to the current supplier, so a single failure no longer stops a product. Every rating below is a company claim from its own pages, not independent proof.

Factor	TTM (replaces Jade)	Infineon (replaces IonPeak)	CeramTec (replaces Orion)	Toray (replaces Meridian)
Technical capability	Strong. Heavy copper (2-12 oz), insulated metal substrates, HDI, flex	Strong. SiC MOSFET and IGBT bare dies, plus modules	Strong. Al_2O_3 , AlN , Si_3N_4 , ZTA substrates, fit for the standard copper-bonding processes used on power modules	Strong. Torayfan polypropylene film, with capacity added for automotive capacitors
Application relevance	Strong. Electrified powertrain, charging stations, energy storage	Strong. Dies for "automotive and industrial needs", with design wins in renewable and industrial	Strong. Industrial drives, e-mobility, renewables	Strong. Film for capacitors in EV motor-inverter circuits
Footprint	Strong. 22 factories (16 North America, 6 Asia Pacific)	Strong. Villach and Dresden (300mm), Kulim in Malaysia (200mm SiC)	Partial. 18 sites in Europe, the USA and Asia, but the substrate plants aren't named	Partial. Only the Tsuchiura plant (Japan) is named
Scale	Strong. 16,000+ staff	Strong. Kulim EUR 2bn first phase, up to EUR 5bn second	Strong. About 3,500 staff	Partial. Capacity up 60% at Tsuchiura, no volume figures
Industry presence	Partial. Automotive 13% of sales, medical/industrial/instrumentation 14%, no named customers	Strong. About EUR 5bn of design wins from six automotive OEMs plus renewable and industrial customers (Aug 2024)	Partial. Sectors listed, no named customers	Partial. Automotive capacitor use stated, no named customers

Factor	TTM (replaces Jade)	Infineon (replaces IonPeak)	CeramTec (replaces Orion)	Toray (replaces Meridian)
Qualification	Partial. IATF 16949 in its factsheet, "TS 16949" (the older name) on its automotive page	Not shown. Certificate page not readable	Not shown. Only RoHS on the pages read	Not shown

Financials (latest public figures we found)

Company	Revenue	Profit signal	Source
TTM	USD 2.44bn (FY2024), 16,000+ staff	Not checked	TTM factsheet, May 2025
Infineon	EUR 14.662bn (FY2025, to 30 Sep 2025), down 2%	Segment Result EUR 2.560bn, 17.5% margin	Infineon press release
CeramTec	About EUR 685m (2025), about 3,500 staff	Not checked	CeramTec company page
Toray	JPY 2,563.3bn (year to March 2025), up 4.0%	Operating income JPY 127.5bn, up 121.1% (about 5% of revenue)	Toray results summary

Our data gives no revenue for the fictional suppliers, so sizes can't be compared directly. A large listed group should carry less financial risk, which matters most for Meridian, whose own finances are the weakest in the network. We have not checked these companies' balance sheets.

Jade to TTM

- **What we replace.** Jade makes program-specific circuit-board substrates for Cobalt's C10 control board and Grove's and HarborSense's B10 battery-monitoring board. DOC-079 says no second source is qualified.
- **Why it works.** TTM makes boards of this kind, including heavy-copper power boards, at plants in North America and Asia, and, on paper, a scale suited to three Tier-1s (16,000+ staff). Real capacity for our volumes is for sourcing to validate.
- **How it works.** TTM would be qualified on both boards in about 12 weeks (indicative figure from our data) and run alongside Jade.
- **Open points.** No TTM page names battery-monitoring boards. Schweizer is the backup: it names battery management and charging stations and holds IATF 16949 at both plants, but it has only 2 plants and about 500 staff.

IonPeak to Infineon

- **What we replace.** The power dies in the power assemblies built by Aster and Boreal. They share a parent and the same IonPeak die family (DOC-075, DOC-078), so a different assembler wouldn't help. The fix has to be an independent die maker.
- **Why it works.** Infineon sells both SiC and IGBT dies and runs three large fabs in different places.
- **How it works.** The modules would be re-qualified with Infineon dies, about 16 weeks (indicative). It is not a drop-in swap.
- **Open points.** Check that Infineon's own inputs don't come from Z01 like IonPeak's (Umber, Verdant). Our data doesn't say whether IonPeak's dies are SiC or IGBT. PolarSwitch is named in DOC-078 only as a prospect.

Orion to CeramTec

- **What we replace.** Ceramic substrates that go into the same Aster and Boreal power assemblies, made in Z01.
- **Why it works.** CeramTec makes the standard ceramic materials for power modules and has 18 sites across three regions.
- **How it works.** A second substrate source qualified on the power assemblies, about 16 weeks (the parent component's figure, used as a stand-in).
- **Open points.** Which plant would make them, and its certificates. Our data doesn't state the ceramic type Orion uses.

Meridian to Toray

- **What we replace.** The dielectric film inside Delta Capacitor Works' C20 DC-link capacitors. Delta is the only C20 supplier, so P1 and P2 (USD 1,144m) depend on this film.
- **Why it works.** Toray makes capacitor film for EV inverters and is a very large group, which suits backing up a financially stressed supplier.
- **How it works.** A second film source qualified at Delta, with about 8 weeks as the indicative figure for C20.
- **Open points.** Only one plant is named on the pages read. Our data doesn't say which film material Meridian makes; polypropylene is our assumption because C20 is a film capacitor.

Links and sources

All pages below were opened on 3 Oct 2026. Page dates are given only where the page shows one.

Company	Page	What it supports	Page date
TTM	Automotive capabilities	Heavy copper 2–12 oz, insulated metal substrates, "TS 16949" sites, named plants	None shown
TTM	Automotive applications	Electrified powertrain, charging stations, energy storage	None shown
TTM	Factsheet	22 factories, 16,000+ staff, USD 2.44bn sales, end-market shares, IATF 16949	May 2025
Infineon	SiC bare dies	CoolSiC bare dies are offered	None shown
Infineon	Bare die product table	SiC and IGBT bare dies, "automotive and industrial needs"	None shown
Infineon	Kulim fab article	Investment of EUR 2bn and up to EUR 5bn, design wins of about EUR 5bn, Villach and Dresden	8 Aug 2024
Infineon	Dresden fab release	Dresden fab, EUR 5bn investment	Not captured
Infineon	FY2025 results release	Revenue EUR 14.662bn, Segment Result EUR 2.560bn	Year to 30 Sep 2025
CeramTec	Ceramic substrates	Al ₂ O ₃ , AlN, Si ₃ N ₄ , ZTA, DCB/AMB, industrial drives and e-mobility	None shown
CeramTec	Power electronics news	Substrates for power electronics, made in Europe	11 Aug 2026
CeramTec	Worldwide sites	German sites plus UK, China, US	None shown

Company	Page	What it supports	Page date
CeramTec	Company facts	About 3,500 staff, about EUR 685m (2025), 18 sites	2025 figures
Toray	Torayfan page	Capacitor applications, first made in Japan	None shown
Toray	Capacity expansion article	60% more capacity at Tsuchiura for automotive capacitor film, EV inverter use	Expansion planned for 2022
Toray	Results summary	Revenue JPY 2,563.3bn, operating income JPY 127.5bn	Year to Mar 2025
Schweizer (backup for Jade)	Sectors	Battery management, e-mobility charging stations	None shown
Schweizer (backup for Jade)	Company facts	2 plants, about 500 staff, IATF 16949 at both	None shown
Schweizer (backup for Jade)	PCB technologies	Heavy copper, inlay boards, embedded chips; no assembly mentioned	None shown

From the case data: the Components Context sheet (lead times and screening criteria), DOC-075 (Aster and Boreal share a parent), DOC-078 (no second die source) and DOC-079 (no independent substrate), plus our risk scorecard.

Limits and next steps

The shortlist is a starting point for sourcing, and these limits belong in the pitch.

- **Claims, not proof.** Every rating comes from the company's own pages. Nothing was checked in an independent source such as a certificate database.
- **Z01 is fictional.** "Outside Z01" can't be tested for a real company, so footprint means plants spread across regions.
- **Scale is unknown.** Our data has no supplier spend or volume. USD 1.56B is NovaDrive's product revenue, not a purchase volume.
- **Lead times are component-level.** 12 weeks (C10, B10), 16 (M10/M20) and 8 (C20) come from the Components Context sheet. Orion's 16 weeks is borrowed from the parent component.

- **Item details are missing.** Our data doesn't say which die type, ceramic or film material the suppliers use, or the board specification. Engineering has to confirm exact fit.
- **Ranking depends on weights.** Jade is 1st under our active weights and 3rd or 4th under the two test weightings, so it stays in the top four either way.

Next steps

1. Check the IATF 16949 certificates for TTM and Infineon, and CeramTec's and Toray's quality pages.
2. Verify the low-confidence names: Umber, Verdant, Warden, Xenon, Yarrow.
3. Give sourcing four questions per alternate: can you make our exact item, which plant would make it, what spare capacity do you have, and how long would qualification really take?
4. Build the pitch deck from the network map, the risk table and this shortlist.